

ORIGINAL ARTICLE

Influence of inhaled corticosteroids on pubertal growth and final height in asthmatic children

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To cite this article: De Leonibus C, Attanasi M, Roze Z, Martin B, Marcovecchio ML, Di Pillo S, Chiarelli F, Mohn A. Influence of inhaled corticosteroids on pubertal growth and final height in asthmatic children. *Pediatr Allergy Immunol* 2016; **00**.

Keywords

asthma; children; growth; inhaled corticosteroid therapy; puberty

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Accepted for publication 21 February 2016

DOI:10.1111/pai.12558

Abstract

Background: Controversial data exist on the possibility that inhaled corticosteroids (ICs) affect growth in children with mild-to-moderate asthma. We assessed whether ICs affect growth and final height (FH) in asthmatic children compared to controls.

Methods: A retrospective study was conducted on 113 asthmatic children compared with 66 control children. Asthmatic children presented with mild-to-moderate asthma and had exclusive ICs. Anthropometric data of four specific time-points were collected for both groups (pre-puberty, onset and late puberty, and FH) and converted to standard deviation scores (SDS). Growth trajectories were assessed as follows: (i) in puberty, using peak height velocity (PHV) and pubertal height gain SDS (PHG-SDS); (ii) until FH achievement, using FH-SDS and FH gain SDS (FHG-SDS). Repeated measurement analysis was performed across longitudinal study visits. A general linear model (GLM) was performed in asthmatic group evaluating the effect of corticosteroid type, treatment duration, and cumulative dose on FH corrected for multiple variables.

Results: At pre-puberty, height and weight SDS were similar between the groups ($p > 0.05$). Height SDS progressively declined over the study period in asthmatic patients from pre-puberty to FH (p -trend < 0.05), whereas it did not change over time in controls (p -trend > 0.05), in both boys and girls. Asthmatic children had exclusive ICs [budesonide ($n = 36$) vs. fluticasone ($n = 43$) vs. mometasone ($n = 34$)] for a mean period of 6.25 ± 1.20 years and a mean cumulative dose of 560.07 ± 76.02 mg. They showed decreased PHG-SDS and lower PHV compared to controls (all $p < 0.05$). FH-SDS and FHG-SDS were significantly reduced in asthmatic group compared to controls. FH in asthmatic patients was 2.5 ± 2.89 cm lower in boys and 2.0 ± 2.03 cm lower in girls than controls. The GLM showed that FH achievement was dependent on the type of ICs, duration of the treatment, and cumulative dose ($p < 0.05$).

Conclusions: ICs affect pubertal growth determining reduced final height in asthmatic children compared to controls, in a dose- and duration-dependent manner.

Asthma is a chronic inflammatory disease of the airways that affects a growing number of children and adolescents (1). Inhaled corticosteroids (ICs) are the mainstay of treatment in persistent asthma, with a stepwise approach to increasing doses of ICs depending on asthma severity and control (2). Current guidelines recommend low-dose ICs as first-line therapy for patients with mild persistent asthma and medium-dose ICs or combination therapy with long-acting beta2-agonists as the

preferred therapy for moderate asthma (2). Inhaled corticosteroid therapy is effective at reducing asthma symptoms, decreasing airway hyper-responsiveness, controlling airway inflammation, reducing the frequency and severity of exacerbations, and asthma mortality (2).

Although the ICs are effective in controlling the disease, there is evidence suggesting that they might be associated with growth-suppressing effects (3). A significant portion of the dose

(60–90%) is deposited in the mouth, absorbed in the gastrointestinal tract, and subject to first-pass metabolism, whereas 10–40% that reaches the airways can cause systemic side effects by being absorbed into the general circulation (3, 4). The inhibitory effect on growth has been shown to be dose-dependent (5) and seems to be related to the suppression of bone formation and inhibition at various levels of the growth hypothalamic-pituitary axis and adrenal function (6, 7). It has also been evidenced that the effect of ICs on linear growth is associated more strongly with the IC molecule than with the absolute dose (8). IC molecules have a different therapeutic index, being related to receptor affinity, pulmonary, and oral bioavailability (9), and they differ even when they are used at the same dose (8).

However, there are conflicting and controversial data in this field, with some reports showing that ICs inhibit growth (10–12), while others do not (13–15). Moreover, the majority of the studies are short-term reports evaluating the effects of ICs on growth measured in weeks to months (3, 6, 11), whereas there is the lack of long-term follow-up studies evaluating growth patterns until the achievement of final height (FH). In a longitudinal study with a follow-up into adulthood, Kelly et al. (11) showed that the pre-pubertal children treated with 400 µg/day of budesonide for average of 4.3 years had a mean reduction in FH of 1.2 cm compared to those who received placebo. Other reports confirmed slower growth rate with a large variability, ranging from 0.2 to 1.8 cm/year (16, 17), while others showed no effect of ICs on final height (14). These discordant findings have been partly explained by the major limitations of the studies in their design and methodology and the lack of a homogeneous population in terms of age, asthma severity, dose, and treatment duration (3). Therefore, the primary aim of this study was to assess longitudinally from pre-pubertal age to the attainment of FH, the effect of inhaled corticosteroid therapy on growth, and final height in children with mild-to-moderate asthma compared to the control population. Furthermore, we evaluated as secondary aim if the type, duration, and dose of IC therapy had effect on final height.

Methods

Study design, subjects, and inclusion criteria

The study was based on a retrospective review of the hard copy archive of the Paediatric Allergy and Endocrine Clinic of the Department of Paediatrics (University of Chieti, Italy) for the period between January 2000 and December 2014. Records of Caucasian children aged between 5 and 7 years, where longitudinal anthropometric data including pubertal staging were available, were selected (Fig. 1).

Diagnosis of asthma was established by a single pediatric respiratory physician according to the Global Initiative for Asthma guidelines (2). Pre-pubertal children with persistent mild-to-moderate asthma with or without rhinitis, receiving for at least 8 months/year a daily IC asthma medication during the study period, were identified. Anti-asthma therapy in this study was defined as ICs [budesonide (BUD), ITALCHIMICI S.p.A., Pomezia (RM), Italy; mometasone furoate (MF),

MSD International GmbH, Singapore; fluticasone propionate (FP), GlaxoSmithKline S.p.A., Verona, Italy] from low to medium dose based on the severity of asthma. In particular, at each follow-up visit, the asthma severity was determined retrospectively from the level of the treatment required to control symptoms and exacerbations (2).

The assessment of the asthma control was based on symptom control (frequency of daytime and nighttime asthma symptoms), reliever use, and activity limitation, and it was evaluated after 3 months from the first visit (2). Regular follow-up was recommended ranging from 1 to 6 months depending on the control of asthma and patient's reliability. Once asthma was controlled for 3 months, a step-down therapy was considered to identify the minimum dose of medication (2).

At every follow-up visit, patients underwent a complete physical examination, including anthropometric parameters and a lung function evaluation by standard flow/volume curve. Furthermore, the quality of inhalation technique was checked by trained nurses at each visit, and the presence or the absence of airway hyper-responsiveness was measured by indirect bronchial provocation tests (mannitol or exercise stimuli) once a year during follow-up visits.

Control group data were taken retrospectively from charts of patients who attended regularly the Endocrine Clinic for endocrine diseases not affecting growth and pubertal development.

The study was approved by the Ethical Committee of the University of Chieti, and it was conducted in compliance with ethical principles based on the Declaration of Helsinki.

Exclusion criteria

Asthmatic children using systemic corticosteroids for more than 2 weeks/year to treat asthma exacerbations and nasal corticosteroids to treat rhinitis were excluded, along with children going into complete remission for asthma. Specifically, asthma remission was considered if the subject denied medications and respiratory symptoms during the preceding year with the absence of airway hyper-responsiveness and the presence of a normal lung function (2).

For controls, pre-pubertal children with any chronic diseases were excluded, along with physical disabilities, or abnormalities in pubertal development, current malnutrition, or born with low birthweight (below 2500 gr). Detailed clinical characteristics of the control group are shown in Table S1.

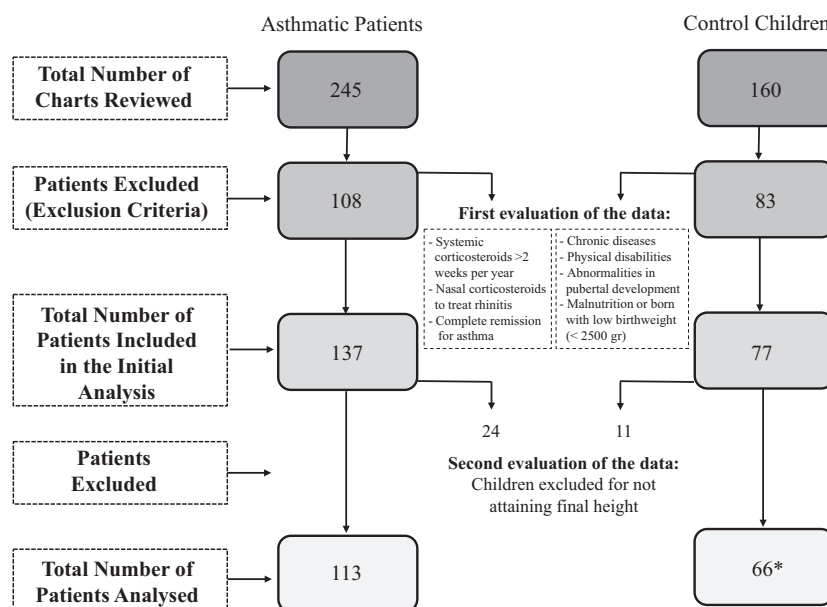
Patients with missing data for anthropometric parameters, pubertal stage, and/or no data on usage of prescribed asthmatic medication were not selected in the study.

Secondly, children not attaining final height were not included in the analysis ($n = 24$ for asthmatics and $n = 11$ for controls) (Fig. 1).

Anthropometric measurements and puberty variables

In our clinical practice, all outpatients are seen by a Paediatric Endocrinologist who performed anthropometric measurements and pubertal staging assessment according to standard methods.

Figure 1 The figure shows the total number of charts reviewed in 14-year period (2000–14). It evidences the number of children excluded in the initial analysis due to the exclusion criteria, and the number of patients excluded in a second analysis, due to the absence of attainment of final height ($n = 24$ for asthmatics and $n = 11$ for controls). The total number included in the final analysis is also shown ($n = 113$ for asthmatics and $n = 66$ for controls).



* Detailed clinical characteristics of the control group are shown in Supplementary Table 1

Height was measured, without shoes, with an Harpenden stadiometer (Holtain, Wales, UK) three times to the nearest 0.1 cm. Weight was measured with the child in light clothing to the nearest 0.1 Kg with a calibrated scale (Salus, Inc., Milan, Italy). Each subjects stood straight with feet placed together and flat on the ground, heels buttock and scapulae against the vertical backboard arm loose and relaxed with the palms facing medially, and the head positioned in the Frankfurt plane. All equipment (scale, stadiometer) was calibrated at the beginning of each study visit.

Pubertal onset and progression were based on direct evaluation of breast development in girls and testicular size in boys according to Tanner's criteria (19, 20).

Pre-puberty was defined as the period before the appearance of pubertal signs (Tanner stage 1). The onset of puberty (Tanner stage 2) was defined for girls as the appearance of breast bud stage and for boys as the enlargement of the scrotum and testes which reached a volume of 4 ml (19, 20). The late puberty (Tanner stage 4) was defined as the appearance of menarche in girls and as the achievement of testicular volume of 25 ml in boys (19, 20).

Spirometric and flowmetric measurements

Lung function was assessed in Paediatric Allergy Clinic of the Department of Pediatrics of Chieti by flow/volume curves (using spirometer Master Screen; Viasys GmbH – Erich Jaeger, Hoechberg, Germany). They were made in the standing position using a nose clip. Until three consecutive technically acceptable curves were achieved according to ATS/ERS guidelines (21). The main spirometric parameters included: FEV₁, forced vital capacity (FVC), FEV₁/FVC, and forced expiratory flows between 25% and 75%.

Data collection

Anthropometric data of four specific time-points were collected for both groups and converted to standard deviation scores (SDS): visit 1 = baseline pre-pubertal visit; visit 2: onset of puberty; visit 3 = late puberty; visit 4 = achievement of final height.

The anthropometric data of pre-puberty were taken from the visit carried out at least 6 months prior to the onset of puberty. The anthropometric data of onset of puberty were taken from the visit carried out at Tanner stage 2. The anthropometric data of late puberty were taken from the visit carried out at Tanner stage 4. Moreover, the anthropometric data of final height were collected, where FH was established from the point at which height growth velocity was less than 1 cm/year after the pubertal growth spurt (22).

For the asthmatic children, the data on detailed residential characteristics, environmental factors, allergic heredity, birth-weight, breastfeeding, and other key exposure factors were obtained from questionnaires to parents administered once a year. At every follow-up visit, similar questionnaires were mailed to the parents, relied on symptoms related to asthma, other allergic diseases, and information on IC and reliever use.

The treatment adherence was assessed by parent reports on IC use. Although the compliance was not measured electronically (23), these data have high concordance with dispersing of asthma medications during a 3- to 6-month window (the time frame for parent reports in this study) (24).

Growth parameters

All height and weight data were adjusted for age and gender by calculating SDS at each time-point visit (visit 1–4), on the basis of reference data for the Italian population (25). Target height

(TH) for children was calculated using parents height by formula (paternal height + maternal height + 13)/2 cm for boys and (paternal height + maternal height - 13)/2 cm for girls and converted to SDS (26). Maternal height and paternal height were measured in the same standardized way as in children. In addition, to take into account, the patient's genetic potential for growth parentally adjusted height SDS (PAH-SDS) at final height was also calculated, which is represented by the difference between the child's final height SDS and target height SDS (27).

Growth trajectories were assessed as follows: (i) in puberty, using age at peak height velocity (APHV) and peak height velocity (PHV), along with pubertal height gain SDS (PHG-SDS), as previously reported (28, 29); (ii) until final height achievement, using FH-SDS and total height gain SDS, as the difference between FH-SDS and height SDS at pre-puberty (27, 28).

For each asthmatic subject, the duration of use of each type of steroid inhaler and the mean daily dose were determined. Secondly, the cumulative dose of inhaled corticosteroids was calculated by multiplying the total amount on inhaled corticosteroid prescribed (or equivalent dose) in milligram by the treatment time and expressed in milligrams, as previously shown (30, 31).

Statistical analysis

Statistical analysis was performed using SPSS version 19.0 software for Windows (SPSS Inc, Chicago, Illinois, USA). All data were expressed as mean $\pm$ SD unless otherwise specified. *p* Values < 0.05 were considered statistically significant.

Differences in continuous variables were assessed by unpaired *t*-test. Differences in categorical variables were assessed by chi-squared test or Fisher's exact test.

Repeated measurement analysis of variance with Bonferroni *post hoc* was applied to assess differences across the longitudinal study visits. A general linear model (GLM) was performed in asthmatic group evaluating the effect of corticosteroid type (budesonide vs. mometasone furoate vs. fluticasone propionate), treatment duration, and cumulative dose on FH corrected for multiple variables known to affect growth (age, gender, weight SDS at final height, and asthma severity). To investigate the effect on FH of different molecules with diverse duration and dosages, treatment duration (years) and cumulative dose (mg) were categorized into three groups based on their distribution in percentiles [(i) $\leq$ 25th percentile; (ii) between the 25th and 75th percentiles; (iii) $\geq$ 75th percentile]. The 25th and 75th percentiles were calculated using SPSS program corresponding to 5.9 and 7.9 years, respectively, for treatment duration, and 400 and 670 mg, respectively, for cumulative dose.

Results

Physical growth over time: intragroup and gender differences

The anthropometric data of asthmatic and control children are shown in Table 1 and in Table S2 for SDS values.

Specifically for the asthmatic patients, clinical characteristics at baseline and under IC treatment are shown in Table 2, for both boys and girls.

At pre-puberty (visit 1), onset of puberty (visit 2), and late puberty (visit 3), the asthmatic and control groups were similar for age, height, and weight SDS for both boys and girls ($p > 0.05$).

The age at achievement of FH was similar between the groups, as well as weight SDS ($p > 0.05$). However, FH-SDS was significantly lower in asthmatic children compared to controls in boys and girls ($p < 0.05$). When assessing final height in centimeters, FH in asthmatic patients was 2.5 ± 2.89 cm significantly lower in boys [mean (SD) 173.55 (3.79) vs. 176.04 (2.37) cm, $p = 0.028$] and 2.0 ± 2.03 cm in girls than controls [161.16 (4.79) vs. 163.44 (3.49) cm, $p = 0.047$] (Table 1). There was a statistically significant decrease in PHG-SDS in asthmatics compared to controls [in boys: -0.21 (0.89) vs. 0.03 (0.41) SDS, $p = 0.036$; in girls: -0.35 (0.93) vs. 0.02 (0.48) SDS, $p = 0.012$] and lower PHV compared to controls [in boys: 6.54 (1.32) vs. 8.18 (1.46) (cm/year), $p = 0.006$; in girls: 5.36 (1.58) vs. 7.10 (1.80) (cm/year), $p = 0.015$], with similar APHV in both boys and girls ($p > 0.05$).

Total height gain SDS [in boys: -0.55 (0.68) vs. 0.03 (0.36) SDS, $p = 0.001$; in girls: -0.56 (0.72) vs. 0.05 (0.48) SDS, $p = 0.002$] and PAH-SDS at final height [in boys: -0.27 (0.93) vs. 0.19 (0.21) SDS, $p = 0.038$; in girls: -0.17 (0.19) vs. 0.10 (0.20) SDS, $p = 0.05$] were significantly reduced in asthmatic children compared with control children. Of note, TH-SDS was similar between the groups ($p > 0.05$) (Table S2).

Over time, physical growth, as assessed by height SDS, was significantly different between the two groups. Whereas in the asthmatic group, mean height SDS progressively and significantly declined over the study period from pre-puberty to FH [from 0.26 (0.82) to -0.29 (0.65) SDS in boys; from 0.40 (0.96) to -0.17 (0.79) SDS in girls, p -trend < 0.05], in the control group, it did not significantly change over time [from 0.16 (0.64) to 0.20 (0.59) SDS in boys; from 0.15 (0.65) to 0.19 (0.55) SDS, p -trend > 0.05] (Fig. 2).

The effect of ICs on final height

Asthmatic children had exclusive ICs [budesonide ($n = 36$) vs. fluticasone ($n = 43$) vs. mometasone ($n = 34$)] for a mean period of 6.25 ± 1.20 years and a mean cumulative dose of 560.07 ± 76.02 mg.

The GLM showed a significant effect of corticosteroid type, duration, and cumulative dose on final height (all $p < 0.05$).

Specifically, children using Fluticasone had significantly lower final height [mean (SD): 164.04 (6.72) cm] compared to children using either budesonide [169.41 (10.47) cm] or mometasone [172.82 (6.98) cm].

Moreover, longer duration of the therapy was significantly associated with lower FH [≤ 5.9 years: 173.44 (5.59) cm; > 5.9 to < 7.9 years: 168.53 (9.70) cm; ≥ 7.9 years: 163.53 (9.35) cm], along with larger cumulative doses associated with reduced FH [≤ 400 mg: 172.98 (6.23) cm; > 400 to < 670 mg: 169.11 (5.42) cm; ≥ 670 mg: 163.52 (8.05) cm] (Fig. 3).

Table 1 Clinical characteristics of the study population by gender. Anthropometric data are not standardized

Clinical characteristics	Boys (n = 86)			Girls (n = 93)		
	Asthmatic (n = 57)	Controls (n = 29)	p	Asthmatic (n = 56)	Controls (n = 37)	p
Pre-pubertal stage						
Age (years)	10.83 ± 1.03	11.09 ± 1.08	0.27	10.16 ± 0.97	10.22 ± 1.04	0.96
Height (cm)	144.51 ± 6.89	145.34 ± 6.45	0.30	141.84 ± 8.91	140.13 ± 7.11	0.33
Weight (Kg)	40.35 ± 6.07	40.57 ± 9.58	0.57	36.65 ± 8.21	36.04 ± 6.61	0.51
Onset of puberty						
Age (years)	11.51 ± 1.06	11.87 ± 1.23	0.20	10.78 ± 1.19	10.84 ± 1.07	0.75
Height (cm)	148.19 ± 8.33	149.96 ± 5.25	0.34	145.65 ± 7.32	144.73 ± 5.92	0.53
Weight (Kg)	42.32 ± 6.91	44.47 ± 7.58	0.33	41.10 ± 6.36	40.14 ± 5.02	0.65
Late puberty						
Age (years)	13.82 ± 1.11	14.07 ± 0.97	0.21	13.38 ± 0.75	13.01 ± 0.79	0.69
Height (cm)	162.91 ± 6.47	168.03 ± 5.05	0.002	158.77 ± 6.46	157.13 ± 3.93	0.35
Weight (Kg)	55.71 ± 7.59	58.02 ± 8.65	0.33	53.10 ± 8.65	51.61 ± 7.38	0.88
Final height						
Age (years)	17.28 ± 0.96	17.37 ± 1.10	0.91	16.87 ± 1.08	16.45 ± 0.72	0.30
Height (cm)	173.55 ± 3.79	176.04 ± 2.37	0.028	161.16 ± 4.79	163.44 ± 3.49	0.047
Weight (Kg)	69.76 ± 5.69	69.75 ± 6.31	0.84	58.84 ± 7.89	56.81 ± 4.74	0.17
Pubertal growth						
Pubertal height gain (cm)	25.36 ± 8.02	26.04 ± 6.38	0.71	15.57 ± 5.24	18.50 ± 4.97	0.034
APHV (year)	12.74 ± 1.09	12.89 ± 1.41	0.63	11.82 ± 1.98	11.60 ± 1.10	0.95
PHV (cm/year)	6.54 ± 1.32	8.18 ± 1.46	0.006	5.36 ± 1.58	7.10 ± 1.80	0.015
Total growth						
Target height (cm)	174.80 ± 3.82	175.03 ± 3.53	0.34	162.26 ± 4.52	163.04 ± 5.09	0.58
Total height gain (cm)	29.00 ± 7.08	30.81 ± 7.22	0.16	19.35 ± 7.69	23.27 ± 6.72	0.003

Data are expressed as mean ± SD; SD, standard deviation; N, number; PHV, peak height velocity; APHV, age at peak height velocity; PAH, parentally adjusted height. The values shown in bold indicate p Values statistically significant (< 0.05).

Table 2 Clinical characteristics of the asthmatic population at baseline and under ICs

Clinical characteristics at baseline	Boys (n = 57)	Girls (n = 56)	p
Asthma severity			
Mild (n)	33	39	0.241
Moderate (n)	24	17	
Allergic rhinitis, n (%)	14/57 (24.5)	11/56 (19.6)	0.801
Parental smoking, n (%)	8/57 (14.0)	7/56 (12.5)	0.889
Clinical characteristics under ICs			
Use of corticosteroids other than ICs			
Nasal steroids (%)	0	0	
Oral steroids (%)	0	0	
Average cumulative IC dose (mg)	561.91 ± 74.01	556.39 ± 84.97	0.710
Compliance (%)	89	91	0.455
No. of exacerbation/year	3.96 ± 1.37	3.75 ± 1.10	0.616

Data are expressed as mean ± SD; SD, standard deviation; No., number; ICs, inhaled corticosteroids.

Discussion

This study showed that inhaled corticosteroid therapy affects pubertal growth determining reduced final height in children with asthma and that the growth-suppression effect is dependent on IC molecule, dose, and duration of the treatment. Our findings on asthma and IC use on height are consistent with other studies reporting that long-term use of ICs for more than 12 months in asthmatic children is associated with reduction in growth velocity and final height (16, 17, 32).

We showed that the IC therapy in asthmatic group induced a greater reduction in FH (2.5 cm in boys and 2.0 cm in girls), compared to the longitudinal study of Kelly et al. (11), reporting a reduction of 1.2 cm for a mean period of 4.3 years in pre-pubertal children treated with 400 µg/day of budesonide. This different finding could be related to the different design of this study (retrospective), to the greater IC dose (560.07 mg), and to the longer period of observation (6.25 years) used in this report. In line with this observation, it has been observed that higher IC doses and longer therapy

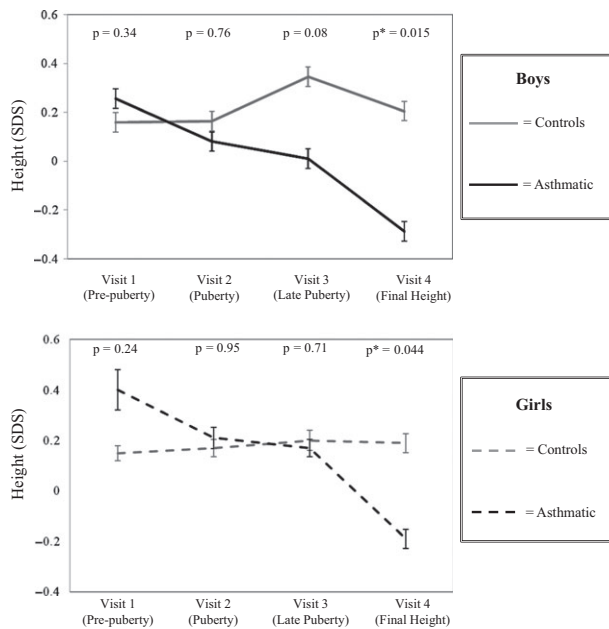


Figure 2 Height SDS across study visits in the asthmatic patients vs. controls by gender. p Value results from differences in height SDS between asthmatic and control children, assessed by unpaired *t*-test. The figure shows data as mean $\pm$ SD (error bars).

are associated with greater reductions in growth velocity (10, 33). Moreover, final height SDS scores of children on ICs were on average 0.5 lower than controls, as previously reported by Littlewood et al. (34). As a strength of this study, we analyzed the genetic potential by the assessment of PAH at final height.

We found that children with asthma did not reach their genetic potential, having a reduced PAH compared to controls at final height.

It has to be considered that chronic diseases such as asthma have growth-suppressing influence effects by itself and that this can confound studies of ICs on growth. Mechanisms underlying this effect remain obscure, but a growth-suppressing influence of endogenous cytokines and glucocorticoids produced in response to illness and inflammation appear likely (35). Of note, we included the severity of asthma as a covariate in the analysis and observed a growth-suppression effect of ICs on final height independently of asthma.

Association between ICs and height during the pubertal years is collectively inconclusive. The pubertal growth phase is variable, as the endogenous GH secretion decreases and the susceptibility to growth suppression from a variety of exogenous influences increases (36). To our knowledge, this is the first study to have compared pubertal phases and characterized pubertal growth spurt in asthmatics vs. controls. Only, a recent longitudinal study has characterized pubertal staging on a population-based cohort of asthmatic patients, identifying no association between asthma, IC use, and pubertal staging (24). The authors collected at ages 1, 2, 4, 8, and 12 years, parent report on asthma, and IC use in the previous 12 months. At 8 and 12 years, height was ascertained at a clinical visit and child-reported, respectively, whereas at 12 years children answered puberty-related questions. The authors concluded that asthma at 8 years was not associated with pubertal staging at 12 years in either sex. Furthermore, they confirmed the effect of IC therapy on growth. However, this study has the limit of lacking data on asthma severity and on dosing or duration of treatment, along with the absence of controls that could explain the different result. Moreover, it relied on self-

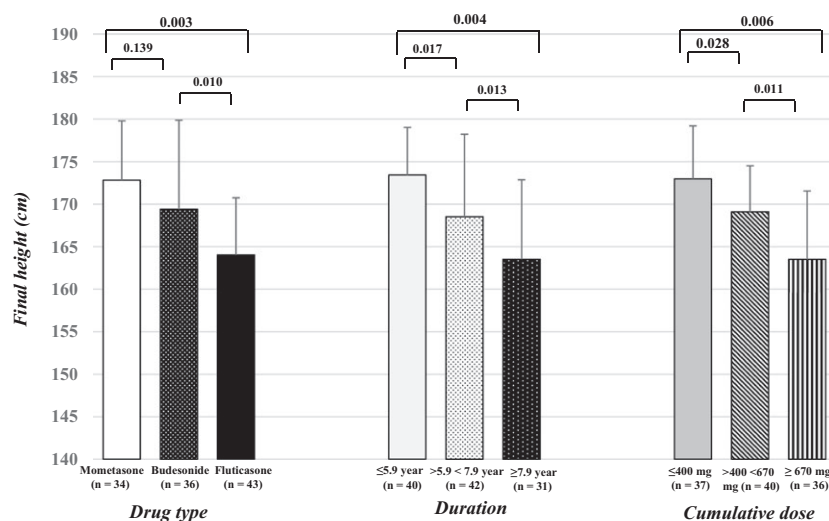


Figure 3 The figure shows the GLM results evaluating the effect of drug molecule, treatment duration, and cumulative dose on adjusted mean final height in centimeter. The GLM has been adjusted for age, gender, asthma severity, and weight SDS at final height. Treatment duration (years) and cumulative dose (mg) have been categorized into three groups based on percentiles distribution of the variables: (i) $\leq$ 25th percentile, (ii) $>$ 25th $<$ 75th percentiles, (iii) $\geq$ 75th percentile. Data are presented as mean values for final height and SD (error bars). p-Values are presented for the comparison between the groups for each variable.

reported height data and did not benefit from the regular evaluation by an endocrinologist who reviewed physical signs associated with Tanner staging. In according to this study, we did not find an association with asthma and pubertal staging in boys and girls. On the other hand, we identified a progressive decline in height SDS from pre-puberty to later stages of pubertal maturation, specifically from late puberty to the achievement of FH. We also showed that the pubertal height gain SDS and peak height velocity were reduced in children with asthma compared to controls, both predictors of lower pubertal growth spurt in asthmatic children in both boys and girls. The explanation of this finding remains speculative, but it may result from the inhibitory effect of exogenous corticosteroids on growth-related pathways, including the inhibition of pituitary GH secretion during childhood along with a combined adrenal suppression during the pubertal growth spurt. The adrenal gland is in fact a critical stimulator of growth during the pubertal spurt, producing sex hormones which markedly stimulate GH secretion (3, 6, 36).

In addition, the systemic effects of ICs on growth seem to be dependent on both dose and duration of IC therapy. The degree of systemic effects is reliant on the pharmacokinetic properties (absorption, distribution, and elimination) that vary among molecules for different characteristics (lipophilicity and protein-binding characteristics). In our study, the effect of ICs on final height was associated with the type of corticosteroid molecule. Fluticasone appeared to cause a significant reduction compared to the budesonide and mometasone groups. Fluticasone has been shown to cause more likely systemic effects compared to the other inhaled corticosteroids (6, 37), and this might depend on the higher lipophilic properties and the longer half-life of the molecule (6, 37). Clark et al. (38) showed that fluticasone administered to asthmatic children via a large volume spacer caused adrenal suppression as measured by urinary cortisol. Budesonide, however, at the same dose did not have this effect, concluding that fluticasone is more potent at inhibiting the hypothalamic–pituitary–adrenal (HPA) axis than budesonide and this suppression is dose-related (38). On the other hand, we identified the lowest effect of mometasone on final height, which has been hypothesized causing a lower

effect on HPA-axis function (39). However, in our study, we did not assess the symptomatic and/or biochemical evidence of HPA-axis suppression by the different IC molecules.

Moreover, in alignment with other reports in this field (11, 40), we showed that higher cumulative dose and greater duration of the treatment had a negative impact on the achievement of final height.

The main strength of the present study is to have reviewed the linear growth until the achievement of final height and assessed the pubertal growth in asthmatic children compared to controls. However, some limitations need to be acknowledged. These include the retrospective design of the study with potential selection bias, the lack of information on bone age and pubertal biochemical markers (i.e., DHEAS, estrogen and testosterone, GH, and IGF-1 levels), a small sample size, and a clinical-based instead of a population-based study design. Another potential limitation to assess dose–response effect on growth resides in the use of total cumulative dose regardless of the molecule, although a separate analysis for the IC molecules was performed showing to have a concomitant effect on FH.

In conclusion, this study underlies the importance to consider potential effects of continuous IC therapy on growth. A close follow-up of childhood growth should be performed, balancing the significance of these effects with the importance of asthma control. Furthermore, the consequences of ICs on growth can be prevented or minimized by selecting proper drugs and adjusting treatment dose. This study suggests that regular use of ICs at low or medium daily doses is associated with linear growth suppression in children with mild-to-moderate persistent asthma.

However, additional studies are needed to better characterize the molecule dependency of growth suppression, particularly with newer molecules (ciclesonide and mometasone), to specify the respective role of the molecule, daily dose, and patient age, and to define the growth suppression of IC therapy over a period of several years in children with persistent asthma. Moreover, new studies are needed to assess which factors are associated with catch-up growth versus persistent growth impairment until adult height in children with asthma.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1 Baseline clinical characteristics of the controls by gender.

Table S2 Standardized anthropometric data of the study population by gender.